

BIOGRAPHICAL SKETCH

NAME: Kachergis, George

eRA COMMONS USER NAME (credential, e.g., agency login): GK1279.2015

POSITION TITLE: Research Scientist

EDUCATION/TRAINING

INSTITUTION AND LOCATION	DEGREE (if applicable)	END DATE MM/YYYY	FIELD OF STUDY
Carleton College, Northfield, MN	BA	06/2007	Computer Science & Cognitive Studies
Indiana University, Bloomington, IN	PHD	12/2012	Cognitive Psychology & Cognitive Science
Leiden University, Leiden, ZH	Postdoctoral Fellow	12/2014	knowledge representation for RoboHow Project
New York University, New York, NY	Postdoctoral Fellow	08/2016	active learning & sensemaking in children

A. Personal Statement

As a Cognitive Psychologist, I am skilled in developing computational and psychometric models of language learning and cognitive development, and in analyzing and managing large datasets. I have developed the expertise required to complete this project of harmonizing the *Wordbank* and *Peekbank* datasets through my time as a research scientist in the lab that started both projects. Prior to that involvement, I served on the faculty at Radboud University in the Department of Artificial Intelligence, and at the Donders Institute. I have conducted basic research on language learning and memory for 17 years, including behavioral and modeling experiments investigating the factors that influence the learning and retention of novel words. In the past five years, I have developed and applied psychometric models to *Wordbank* to investigate bilingualism[1], demographic bias[3], and to create more efficient tests of early children's early language[4]. My research seeks to document individual differences in learning environments and outcomes, to increase the diversity of the populations we study, and to improve our measures and theories of language development. I am excited to be working on this new project with highly-qualified colleagues.

1. Tamis-LeMonda CS, Kachergis G, Masek LR, Gonzalez SL, Soska KC, Herzberg O, Xu M, Adolph KE, Gilmore RO, Bornstein MH, Casasola M, Fausey CM, Frank MC, Goldin-Meadow S, Gros-Louis J, Hirsh-Pasek K, Iverson J, Lew-Williams C, MacWhinney B, Marchman VA, Naigles L, Namy L, Perry LK, Rowe M, Sheya A, Soderstrom M, Song L, Walle E, Warlaumont AS, Yoshida H, Yu C, Yurovsky D. Comparing apples to manzanas and oranges to naranjas: A new measure of English-Spanish vocabulary for dual language learners. *Infancy*. 2024 May-Jun;29(3):302-326. PubMed Central PMCID: PMC11019594.
2. Zettersten M, Yurovsky D, Xu TL, Uner S, Tsui ASM, Schneider RM, Saleh AN, Meylan SC, Marchman VA, Mankewitz J, MacDonald K, Long B, Lewis M, Kachergis G, Handa K, deMayo B, Carstensen A, Braginsky M, Boyce V, Bhatt NS, Bergey CA, Frank MC. Peekbank: An open, large-scale repository for developmental eye-tracking data of children's word recognition. *Behav Res Methods*. 2023 Aug;55(5):2485-2500. PubMed Central PMCID: PMC9950292.
3. Kachergis G, Francis N, Frank MC. Estimating Demographic Bias on Tests of Children's Early Vocabulary. *Top Cogn Sci*. 2023 Apr;15(2):303-314. PubMed PMID: 36479833.
4. Kachergis G, Marchman VA, Dale PS, Mankewitz J, Frank MC. Online Computerized Adaptive Tests of Children's Vocabulary Development in English and Mexican Spanish. *J Speech Lang Hear Res*. 2022 Jun 8;65(6):2288-2308. PubMed Central PMCID: PMC9567402.

B. Positions, Scientific Appointments and Honors

Positions and Scientific Appointments

2024 -	Research Scientist, Arizona State University, Cognition, Behavior & Information Unit, Computational Cognitive Models Lab, Phoenix, AZ
2018 - 2024	Research Scientist, Stanford University, Department of Psychology, Language & Cognition Lab, Stanford, CA
2016 - 2018	Assistant Professor of Artificial Intelligence, Radboud University, Donders Institute, Nijmegen

Honors

2022	Diversity and Social Inequality Award, Cognitive Science Society
2015	Irving J. Saltzman Award for Outstanding Graduate Achievement, Indiana University Psychology Department
2013	Robert J. Glushko Dissertation Prize for outstanding interdisciplinary work, Cognitive Science Society
2012	Marr Prize for Best Student Paper, Cognitive Science Society

C. Contribution to Science

1. Efficient, valid, fair, and open assessment of children's early language.

My recent work has leveraged large-scale, open datasets such as *Wordbank* to develop innovative assessment tools that are both efficient and fair. For example, I have created a computerized adaptive test (CAT) version of the Spanish and English Child Development Inventories (CDIs), reducing the assessment from 680 items to approximately 25 without sacrificing validity. These instruments are now included in the NIH-funded Baby Toolbox. Additionally, I have conducted research to identify and mitigate demographic biases in the CDIs, ensuring more equitable measures across diverse populations. My psychometric modeling efforts have also resulted in a novel bilingual CDI tool that better assesses Spanish-English bilingual children. Importantly, all of these tools and models are openly available to promote transparency and accessibility in research and clinical applications.

- a. Kachergis G, Marchman VA, Dale PS, Mankewitz J, Frank MC. Online Computerized Adaptive Tests of Children's Vocabulary Development in English and Mexican Spanish. *J Speech Lang Hear Res.* 2022 Jun 8;65(6):2288-2308. PubMed Central PMCID: PMC9567402.
- b. Tamis-LeMonda CS, Kachergis G, Masek LR, Gonzalez SL, Soska KC, Herzberg O, Xu M, Adolph KE, Gilmore RO, Bornstein MH, Casasola M, Fausey CM, Frank MC, Goldin-Meadow S, Gros-Louis J, Hirsh-Pasek K, Iverson J, Lew-Williams C, MacWhinney B, Marchman VA, Naigles L, Namy L, Perry LK, Rowe M, Sheya A, Soderstrom M, Song L, Walle E, Warlaumont AS, Yoshida H, Yu C, Yurovsky D. Comparing apples to manzanas and oranges to naranjas: A new measure of English-Spanish vocabulary for dual language learners. *Infancy.* 2024 May-Jun;29(3):302-326. PubMed Central PMCID: PMC11019594.
- c. Kachergis G, Francis N, Frank MC. Estimating Demographic Bias on Tests of Children's Early Vocabulary. *Top Cogn Sci.* 2023 Apr;15(2):303-314. PubMed PMID: 36479833.

2. Leading and contributing to developmental open science initiatives

I have been a major contributor in several open science initiatives, from leading analysis and training on multiple ManyBabies projects (multi-lab developmental psychology replication studies) to designing an open-source head-mounted camera for recording infants' egocentric perspective. Critically, I have helped lead the creation and expansion of *Peekbank*, a large-scale, open repository for eye-tracking data that captures real-time language comprehension in young children. By standardizing and harmonizing these diverse datasets, my work has enabled more comprehensive analyses of language processing efficiency and its relationship to language outcomes. This repository supports reproducible research and provides a foundation for testing theories of language development at an unprecedented scale.

- a. Tsui, A. S. M., Carstensen, A., Kachergis, G., ..., Frank, M. C. (registered report accepted pending data collection). Exploring variation in infants' preference for infant-directed speech: Evidence from a multi-site study in Africa. *Dev Sci*.
- b. Visser, I., Geambasu, A., Bergmann, C., Byers-Heinlein, K., Doyle, F. L., Hannon, E., Johnson, S., Kachergis, G., Kosie, J., ... and Levelt, C. (registered report accepted pending data collection). ManyBabies3: A multi-lab study of infant algebraic rule learning. *Dev Sci*.
- c. Zettersten M, Yurovsky D, Xu TL, Uner S, Tsui ASM, Schneider RM, Saleh AN, Meylan SC, Marchman VA, Mankewitz J, MacDonald K, Long B, Lewis M, Kachergis G, Handa K, deMayo B, Carstensen A, Braginsky M, Boyce V, Bhatt NS, Bergey CA, Frank MC. Peekbank: An open, large-scale repository for developmental eye-tracking data of children's word recognition. *Behav Res Methods*. 2023 Aug;55(5):2485-2500. PubMed Central PMCID: PMC9950292.
- d. Long, B. Goodin, S., Kachergis, G., Marchman, V. A., Radwan, S., Sparks, R., Xiang, V., Zhuang, C., Hsu, O., Newman, B., Yamins, D. L. K., Frank, M. C. (2023). The BabyView camera: Designing a new head-mounted camera to capture children's early social and visual environment. *Behav Res Methods*. 2024 Apr;56(4):3523-3534. PMID: 37656342.

3. Computational models of language learning and memory

Throughout my career, I have developed and tested computational models that simulate word learning and memory processes in order to account for empirical data, typically from experiments I design and conduct. My research on building models of word learning has explored the cognitive mechanisms that underpin language acquisition, shedding light on how factors such as repetition, semantic context, and prior knowledge influence vocabulary development. These models have informed both theoretical frameworks and practical approaches to language education and intervention.

- a. Kachergis, G., Yu, C. & Shiffrin, R.M. An associative model of adaptive inference for learning word-referent mappings. *Psychon Bull Rev*. 2012 Apr;19(2):317-24. PMID: 22215466.
- b. Kachergis G, Yu C, Shiffrin RM. A Bootstrapping Model of Frequency and Context Effects in Word Learning. *Cogn Sci*. 2017 Apr;41(3):590-622. PubMed Central PMCID: PMC7683211.
- c. Kachergis, G., Marchman, V. A., & Frank, M. C. Toward a "Standard Model" of Early Language Learning. *Cur Dir in Psych Sci*. 2022 31(1), 20-27. doi: 10.1177/09637214211057836
- d. Kachergis G, Rhodes M, Gureckis T. Desirable difficulties during the development of active inquiry skills. *Cognition*. 2017 Sep;166:407-417. PMID: 28622614.